

Example 12. Consider the sequence $x_n = n$. We claim that this sequence does not converge.

Assume, on the contrary, that (x_n) converges to some $x \in \mathbb{R}$. By Remark 1, following the definition of convergence, the interval

$$(x - 1, x + 1)$$

must contain all but finitely many terms of the sequence. Hence, there exists $N \in \mathbb{N}$ such that

$$n = x_n \in (x - 1, x + 1), \quad \forall n \geq N.$$

In particular,

$$n < x + 1, \quad \forall n \geq N.$$

Thus, the set $\mathbb{N}$ is bounded above by $\max\{N, x + 1\}$, contradicting the fact that $\mathbb{N}$ is not bounded above. Therefore, (x_n) does not converge.

Example 13. Consider the sequence $x_n = (-1)^n$. Observe that the intervals

$$\left(-\frac{3}{2}, -\frac{1}{2}\right) \quad \text{and} \quad \left(\frac{1}{2}, \frac{3}{2}\right)$$

each contain infinitely many terms of the sequence. Hence, no interval containing a possible limit can contain all but finitely many terms of the sequence. Therefore, $((-1)^n)$ is not convergent.

Converge

$$\lim_{n \rightarrow \infty} x_n = x \text{ if } (\forall \varepsilon > 0) (\exists N \in \mathbb{N}) (\forall n \geq N) (|x - x_n| < \varepsilon).$$

Diverge

$$(\forall x \in \mathbb{R}) (\exists \varepsilon > 0) (\forall N \in \mathbb{N}) (\exists n \geq N) (|x_n - x| \geq \varepsilon).$$

A sequence that is not convergent is called *divergent*. This does *not* mean that the terms of the sequence must tend to infinity. A divergent sequence may oscillate, as in the sequence $((-1)^n)$, or it may become arbitrarily large, as in the sequence (n) . Thus, there are different ways in which a sequence can fail to converge. To prove that a sequence is convergent, we must identify its limit and verify the definition of convergence. On the other hand, to prove that a sequence is divergent, it is enough to show that no real number can be its limit.

2.2 Properties of convergent sequences

Definition 9. A sequence (x_n) of real numbers is said to be bounded if there exists a constant $C > 0$ such that

$$|x_n| \leq C, \quad \forall n \in \mathbb{N}.$$

Example 14. The following are some examples of bounded and unbounded sequences.

1. Every constant sequence is bounded.
2. The sequence $(\frac{1}{n})$ is bounded.
3. The sequence $((-1)^n)$ is bounded.
4. The sequence $(\frac{n^2+1}{n^2})$ is bounded.
5. The sequence $(\sin n)$ is bounded.

Definition 10. A sequence (x_n) of real numbers is said to be unbounded if, for every $C > 0$, there exists $n \in \mathbb{N}$ such that $|x_n| > C$.

Example 15. The sequences

$$(n^2), \quad (n^n), \quad (n)$$

are unbounded.

Theorem 6. Every convergent sequence of real numbers is bounded.

Proof. Let (x_n) be a sequence converging to $x \in \mathbb{R}$. Choose $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < 1, \quad \forall n \geq N.$$

By the triangle inequality,

$$|x_n| = |x_n - x + x| \leq |x_n - x| + |x| < 1 + |x|, \quad \forall n \geq N.$$

Now let

$$C = \max\{|x_1|, \dots, |x_{N-1}|, 1 + |x|\}.$$

Then

$$|x_n| \leq C, \quad \forall n \in \mathbb{N}.$$

Hence, (x_n) is bounded. □

The converse of the above result is not true. Can you recall an example?

Theorem 7. Let (x_n) be a sequence of real numbers such that

$$\lim_{n \rightarrow \infty} x_n = x,$$

where $x \neq 0$. Then there exists $N \in \mathbb{N}$ such that

$$|x_n| > \frac{|x|}{2}, \quad \forall n \geq N.$$

Proof. Choose

$$\varepsilon = \frac{|x|}{2} > 0.$$

Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{|x|}{2}, \quad \forall n \geq N.$$

By the triangle inequality,

$$|x| = |x - x_n + x_n| \leq |x - x_n| + |x_n| < \frac{|x|}{2} + |x_n|.$$

Hence,

$$|x_n| > \frac{|x|}{2}, \quad \forall n \geq N.$$

This completes the proof. □

Theorem 8. Let (a_n) be a sequence of real numbers such that

$$\lim_{n \rightarrow \infty} a_n = a.$$

If $a_n \geq 0$ for all $n \in \mathbb{N}$, then $a \geq 0$.

Proof. Suppose, on the contrary, that $a < 0$. Then there exists $\delta > 0$ such that

$$a + \delta < 0.$$

Since $a_n \rightarrow a$, there exists $N \in \mathbb{N}$ such that

$$a_n \in (a - \delta, a + \delta), \quad \forall n \geq N.$$

Since $a + \delta < 0$, every point in the interval $(a - \delta, a + \delta)$ is negative. Thus,

$$a_n < 0, \quad \forall n \geq N,$$

which contradicts the assumption that $a_n \geq 0$ for all $n \in \mathbb{N}$. Therefore, $a \geq 0$. □

Exercise 9. Let (u_n) and (v_n) be two convergent sequences such that

$$u_n \leq v_n, \quad \forall n \in \mathbb{N}.$$

Show that

$$\lim_{n \rightarrow \infty} u_n \leq \lim_{n \rightarrow \infty} v_n.$$

Given two sequences (x_n) and (y_n) , we can construct new sequences by adding, multiplying, or multiplying by a real number. A natural question is: if the original sequences are convergent, what can we say about these newly constructed sequences? In this section, we show that convergence is preserved under these basic algebraic operations.

Theorem 9 (Algebra of Convergent Sequences). Let (x_n) and (y_n) be sequences of real numbers such that $\lim_{n \rightarrow \infty} x_n = x$, $\lim_{n \rightarrow \infty} y_n = y$, and let $\alpha \in \mathbb{R}$. Then:

1. $\lim_{n \rightarrow \infty} (x_n + y_n) = x + y$.
2. $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x$.
3. $\lim_{n \rightarrow \infty} (x_n y_n) = xy$.
4. If $x_n > 0$ for all $n \in \mathbb{N}$ and $x > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{x}$.

Proof. Let $\varepsilon > 0$ be given.

(1) Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2}, \quad \forall n \geq N_1.$$

Similarly, since $y_n \rightarrow y$, there exists $N_2 \in \mathbb{N}$ such that

$$|y_n - y| < \frac{\varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence,

$$\lim_{n \rightarrow \infty} (x_n + y_n) = x + y.$$

(2) If $\alpha = 0$, then the result is obvious. Assume $\alpha \neq 0$. Since $x_n \rightarrow x$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{|\alpha|}, \quad \forall n \geq N.$$

Hence,

$$|\alpha x_n - \alpha x| = |\alpha||x_n - x| < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x.$$

(3) Since $x_n \rightarrow x$, there exists $N_1 \in \mathbb{N}$ such that

$$|x_n| - |x| \leq ||x_n| - |x|| \leq |x_n - x| < 1, \quad \forall n \geq N_1.$$

Also, there exist $N_2, N_3 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{\varepsilon}{2(1 + |y|)}$$

for all $n \geq N_2$, and

$$|y_n - y| < \frac{\varepsilon}{2(1 + |x|)}$$

for all $n \geq N_3$. Let $N = \max\{N_1, N_2, N_3\}$. Then

$$\begin{aligned} |x_n y_n - xy| &= |x_n y_n - x_n y + x_n y - xy| \\ &\leq |x_n||y_n - y| + |y||x_n - x| \\ &\leq (|x| + 1) \frac{\varepsilon}{2(1 + |x|)} + |y| \frac{\varepsilon}{2(1 + |y|)} < \varepsilon. \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} (x_n y_n) = xy.$$

(4) Since $x > 0$, by the previous Theorem 7 there exists $N_1 \in \mathbb{N}$ such that

$$x_n > \frac{x}{2}, \quad \forall n \geq N_1.$$

Also, there exists $N_2 \in \mathbb{N}$ such that

$$|x_n - x| < \frac{x^2 \varepsilon}{2}, \quad \forall n \geq N_2.$$

Let $N = \max\{N_1, N_2\}$. Then, for every $n \geq N$,

$$\left| \frac{1}{x_n} - \frac{1}{x} \right| = \frac{|x_n - x|}{x x_n} \leq \frac{2}{x^2} |x_n - x| < \varepsilon.$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{1}{x_n} = \frac{1}{x}.$$

□

Exercise 10. In Part (4) of the above theorem, the assumption $x > 0$ is essential. What can go wrong if $x = 0$? Give an example to show that the conclusion need not hold.

2.3 Cauchy sequences

So far, every time we proved that a sequence converges, we first guessed its limit and then verified the definition of convergence. This naturally raises the following question:

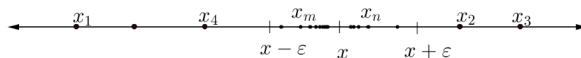
Can we decide whether a sequence is convergent without knowing its limit?

Suppose that (x_n) converges to x . Let $\varepsilon > 0$ be given. Then there exists $N \in \mathbb{N}$ such that

$$|x_n - x| < \varepsilon, \quad \forall n \geq N.$$

Hence, for any $m, n \geq N$, the triangle inequality gives

$$|x_n - x_m| \leq |x_n - x| + |x_m - x| < \varepsilon + \varepsilon = 2\varepsilon.$$



Thus, when the indices are sufficiently large, the terms of a convergent sequence become arbitrarily close to each other. This observation motivates the following definition.

Definition 11 (Cauchy Sequence). A sequence (x_n) of real numbers is said to be Cauchy if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|x_n - x_m| < \varepsilon, \quad \forall m, n \geq N.$$

Example 16. Every convergent sequence is a Cauchy sequence. Is the converse true?

Theorem 10. [Cauchy Criterion] A sequence of real numbers is Cauchy if and only if it is convergent.

Proof. We have already seen that every convergent sequence is Cauchy. We now prove the converse.

Let $E = \{x \in \mathbb{R} : \exists N \in \mathbb{N} \text{ such that } x < x_n, \forall n \geq N\}$.

Let $\delta > 0$. Since (x_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|x_m - x_n| < \delta$ for all

$m, n \geq N_0$. In particular, $|x_{N_0} - x_n| < \delta$ for all $n \geq N_0$. Hence, $x_n \in (x_{N_0} - \delta, x_{N_0} + \delta)$ for all $n \geq N_0$. In particular, $x_n < x_{N_0} + \delta$ for all $n \geq N_0$.

Claim 1. $x_{N_0} - \delta \in E$.

Indeed, taking $N = N_0$, we have $x_n > x_{N_0} - \delta$ for all $n \geq N_0$.

Claim 2. $x_{N_0} + \delta$ is an upper bound of E .

Suppose not. Then there exists $x \in E$ such that $x > x_{N_0} + \delta$. Since $x \in E$, there exists $N \in \mathbb{N}$ such that $x_n > x$ for all $n \geq N$. Hence, for all $n \geq \max\{N, N_0\}$,

$$x_n > x > x_{N_0} + \delta,$$

which contradicts the inequality $x_n < x_{N_0} + \delta$ for all $n \geq N_0$.

Thus, E is nonempty and bounded above. By the completeness of $\mathbb{R}$, let $\ell = \sup E$.

Claim 3. $x_n \rightarrow \ell$.

Let $\varepsilon > 0$. Since (x_n) is Cauchy, there exists $N_0 \in \mathbb{N}$ such that $|x_n - x_{N_0}| < \varepsilon/2$ for all $n \geq N_0$.

By Claim 1, $x_{N_0} - \varepsilon/2 \in E$, and hence $x_{N_0} - \varepsilon/2 \leq \ell$. By Claim 2, $\ell \leq x_{N_0} + \varepsilon/2$. Therefore, $|x_{N_0} - \ell| \leq \varepsilon/2$.

Now, for every $n \geq N_0$,

$$|x_n - \ell| \leq |x_n - x_{N_0}| + |x_{N_0} - \ell| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, $x_n \rightarrow \ell$. □

Example 17. The sequence $(\frac{1}{n})$ is a Cauchy sequence.

Proof. Let $\varepsilon > 0$ be given. Choose $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Then, for any $m, n \geq N$, without loss of generality, assume $m \leq n$. Hence,

$$\left| \frac{1}{m} - \frac{1}{n} \right| = \frac{1}{m} - \frac{1}{n} < \frac{1}{m} \leq \frac{1}{N} < \varepsilon.$$

Therefore, $(\frac{1}{n})$ is a Cauchy sequence. □